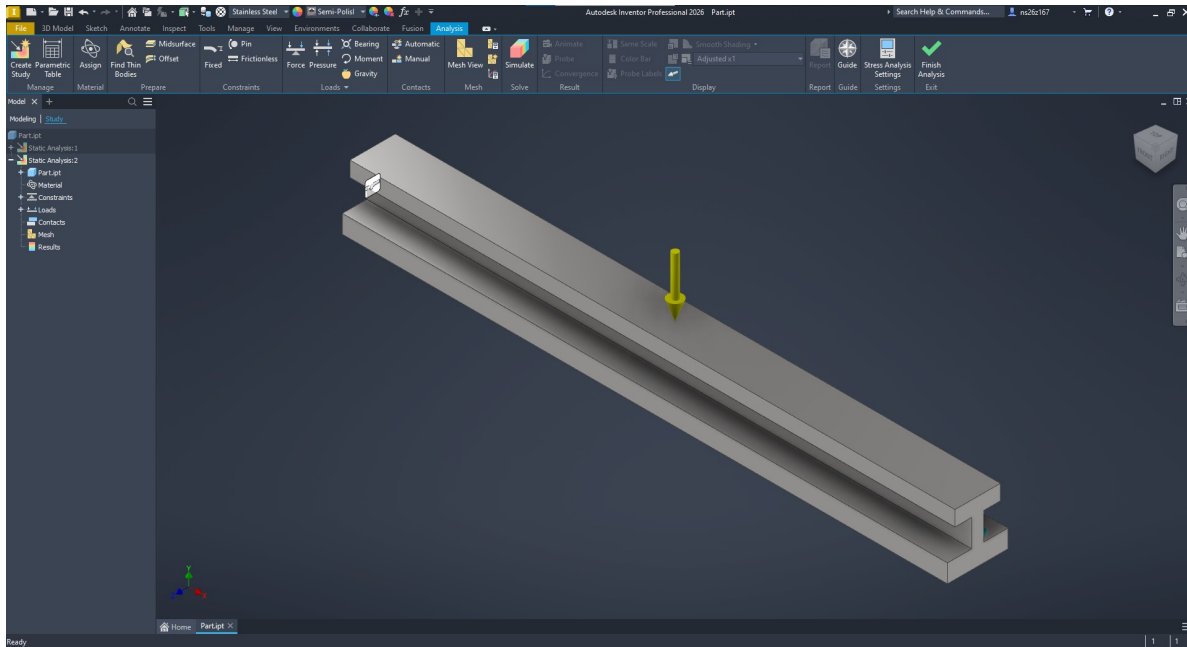


ED 6002: Optimization Methods in Engineering Design (2025-26)

Assignment 1 / Pranav D / ns26z167

At a Glance

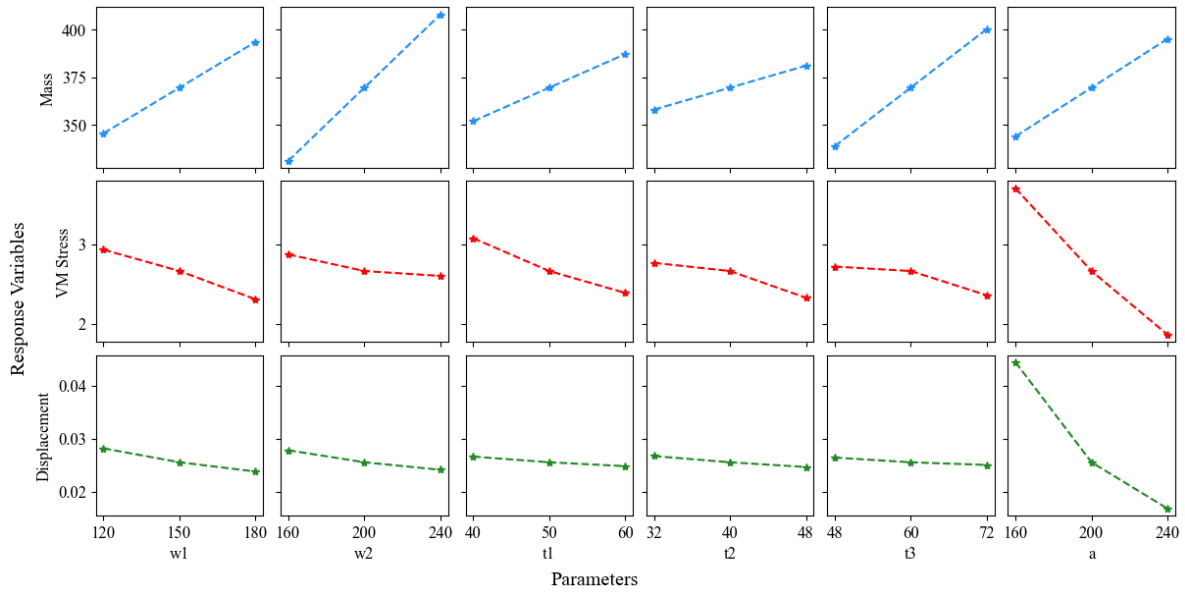
- Created a 2m beam with the specified I cross section
- Applied 9810 N load on the top face
- Simulated for the exhaustive set of parametric configurations
- $3 \text{ levels} \wedge 6 \text{ parameters}$ results in 729 models!



Q1. Sensitivity of each variable on outcomes

- To analyse the sensitivity of each variable in the cross section dimension, I chose to go by OFAT (*One Factor at a Time*) approach.
- Change one variable, keep everything else the same (base dimensions), and analyse the trends in outcomes.

Sensitivity plot



Comments

- Mass is directly proportional to all the parameters – Greater the cross section, greater the mass.
- Though Mass is sensitive to all the parameters, its strongly influenced by $w2$.
- VM stress decrease as size increases, with a contributing for significant reduction compared to others.
- Similarly displacement also decreases with increase in cross section parameters, but again a have higher effect.

Q2. Optima based on DoE

Objective Function:

$$\text{Minimize} : Mass(p)$$